Haoming (Harvey) Chen

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Research Interests: Graph machine learning, generative modeling, flow and diffusion models, network inference, and computer vision

Education

Rice University **Houston, TX**
B.S. in Electrical and Computer Engineering; GPA: 3.98/4.00 *Expected May 2027*
Selected Coursework: Honors Linear Algebra; Machine Learning: Concepts and Techniques; Introduction to Computer Vision; Learning from Sensor Data; Random Signals.

Honors & Awards

- **Louise J. Walsh Scholarship**, Rice School of Engineering, 2025-2026 ($1,500 merit award based on academic record).
- **President's Honor Roll**, Rice University, Fall 2023 and Spring 2025.
- **Member, IEEE-Eta Kappa Nu (HKN)**, the electrical and computer engineering honor society, Theta Rho Chapter; inducted 2026.

Publications

Harvey Chen, Nicolas Zilberstein, and Santiago Segarra. *Prior-Informed Flow Matching for Graph Reconstruction*. *Transactions on Machine Learning Research* (TMLR), 2026. [Paper] [arXiv] [Code]

Harvey Chen, Nicolas Zilberstein, Santiago Paternain, and Santiago Segarra. *Feasible Flow Matching for Graph Reconstruction via Within-Sampling Primal-Dual Guidance*. arXiv preprint arXiv:2609.32980, 2026. [Paper]

Research Experience

Research Assistant **May 2025 - Present**
Rice University, Advisor: Prof. Santiago Segarra *Houston, TX*

- Developed Prior-Informed Flow Matching (PIFM), a rectified-flow generative framework for missing-edge graph reconstruction conditioned on structural priors such as Node2Vec and GraphSAGE.
- Developed and maintained a PyTorch-based research codebase for graph reconstruction and link-prediction tasks across diverse graph datasets; evaluated link-prediction performance and graph-distribution metrics against baseline methods.
- Led experiments, analysis, figure preparation, and writing for a first-author paper published in TMLR.
- Developed a constraint-aware version of PIFM using primal-dual guidance during model inference (arXiv preprint).

Undergraduate Researcher, Trust & Responsibility Summer Research Program **May 2026 - Present**
Rice University, Advisor: Prof. César A. Uribe *Houston, TX*

- Developing an occupation translator that represents occupations as task graphs and aligns taxonomies across time and systems using task embeddings, clustering, subgraph construction, and fused Gromov-Wasserstein optimal transport.
- Built an extraction and embedding pipeline for O*NET releases and ESCO/ISCO taxonomies to examine occupation splits, merges, task reorganization, and ambiguous matches.

Presentations

Harvey Chen. *Prior-Informed Flow Matching for Graph Reconstruction*. Oral presentation, Gulf Coast Undergraduate Research Symposium (GCURS), Rice University, Houston, TX, Oct. 18, 2025.

Teaching Experience

Teaching Assistant **Spring 2026 - Present**
Rice University, Department of Electrical and Computer Engineering *Houston, TX*

- ELEC 241: Fundamentals of Electrical Engineering I (Fall 2026)
- ELEC 242: Signals, Systems, and Transforms (Spring 2026)
- ELEC 378: Machine Learning: Concepts and Techniques (Spring 2026)

Selected Projects

Speech Emotion Classification **Spring 2025**

- Built and evaluated a machine-learning model that classifies emotions from short voice clips; the team achieved the highest classification accuracy among course groups.

NeuralSeed Real-Time Guitar Pedal *[Repository]* **Spring 2026**

- Integrated an RTNeural GRU model into C++ firmware for a Daisy Seed embedded audio platform, enabling real-time neural amplifier and effects processing.
- Ported machine-learning and digital signal-processing components to the board and developed the project website for the team demonstration.

Mentoring & Service

Rice Catalyst Eureka Outreach Program Mentor **Nov. 2024 - Sept. 2026**
Rice University *Houston, TX*

- Mentor Houston-area high-school students on research design, literature reviews, technical writing, and oral presentations; advise student submissions to the Rice Eureka research journal.

Technical Skills

Programming: Python, Java, Go, HTML/CSS, TypeScript, C#, C

Machine Learning and Data: PyTorch, PyTorch Geometric, scikit-learn, NumPy, pandas, SciPy, NetworkX, POT, Matplotlib

Tools: Git, Linux, LaTeX, Unity